

Cloud Migration Strategy

Hybrid Cloud Migration Plan — AWS & GCP | Version 1.0 | January 2026

1. Executive Summary

TechViet Solutions is undertaking a comprehensive cloud migration initiative to transition our entire on-premises infrastructure to a hybrid cloud architecture leveraging Amazon Web Services (AWS) as the primary cloud provider and Google Cloud Platform (GCP) as a secondary provider for specific AI/ML workloads. This migration is driven by several critical business factors: our current data center lease in District 7, Ho Chi Minh City expires in September 2027, and renewal costs have increased by 340% due to the building's reclassification as premium commercial space. Additionally, our on-premises infrastructure has reached 87% capacity utilization, with peak loads during Q4 sales events regularly exceeding 95%, resulting in degraded performance for our flagship ChatAssist product. The total cost of ownership analysis conducted by the Finance team, in collaboration with AWS solution architects, projects a 42% reduction in infrastructure costs over a 5-year period when comparing the current on-premises model (including the new lease terms) against the proposed hybrid cloud architecture. However, this projection assumes successful implementation of reserved instance commitments, spot instance utilization for batch workloads, and the adoption of serverless architectures for at least 30% of our microservices by the end of Year 2. The migration will be executed in four phases over 18 months, beginning with non-critical internal tools in Phase 1 and culminating with the migration of our production ChatAssist platform and customer-facing APIs in Phase 4. Each phase includes a mandatory 30-day parallel-run period where both on-premises and cloud systems operate simultaneously to validate data consistency, performance parity, and failover capabilities.

The risk assessment conducted by the Security and Engineering teams identified 47 distinct risk factors across five categories: data sovereignty (12 risks), security posture changes (15 risks), operational continuity (8 risks), cost management (7 risks), and talent readiness (5 risks). Of these, 8 were classified as Critical, requiring mitigation plans approved by the CTO and CISO before Phase 1 can commence. The most significant critical risk is the potential for data residency violations, as Vietnamese regulations under Decree 13/2023/ND-CP require that certain categories of personal data must be stored within Vietnam's borders. AWS launched its Hanoi region (ap-southeast-5) in Q3 2025, which addresses this requirement, but our architecture must ensure that data routing rules are enforced at the application layer, not just at the infrastructure level. The Data Privacy team, led by Tran Minh Duc (DPO), has developed a data classification matrix that maps each data category to its required storage jurisdiction, and this matrix must be integrated into our CI/CD pipeline as an automated compliance check.

2. Current Infrastructure Assessment

2.1 Data Center Overview

Our primary data center is located at TechPark Tower, Level B2, 1234 Nguyen Van Linh Boulevard, District 7, Ho Chi Minh City. The facility occupies 280 square meters of raised-floor space with N+1 power redundancy and dual-feed cooling systems. The data center was originally provisioned in 2019 when TechViet had approximately 50 employees and operated a monolithic application architecture. Since then,

our infrastructure has grown organically to support 350 employees, 12 microservices, 3 customer-facing products, and over 2 million monthly active users. The following inventory represents our current hardware assets as of the Q1 2026 audit:

Asset Category	Count	Avg Age	Utilization	Replacement Cost
Application Servers (Dell PowerEdge R750)	24	2.8 years	78%	\$8,400 each
Database Servers (Dell PowerEdge R940xa)	6	1.5 years	91%	\$24,000 each
GPU Servers (NVIDIA DGX A100)	4	1.2 years	96%	\$199,000 each
Storage (NetApp FAS8700)	2 clusters	3.1 years	84%	\$180,000 each
Network Switches (Cisco Nexus 9300)	8	2.5 years	62%	\$12,000 each
Load Balancers (F5 BIG-IP i5800)	2	3.8 years	71%	\$45,000 each
Firewall (Palo Alto PA-5260)	2 (HA pair)	2.0 years	45%	\$85,000 each

2.2 Network Architecture and Dependencies

The current network topology follows a traditional three-tier architecture with core, distribution, and access layers. External traffic enters through two redundant 10 Gbps fiber connections from VNPT and Viettel, terminating at the Palo Alto firewall pair configured in active-passive high availability mode. Behind the firewall, the F5 load balancers distribute traffic across the application server farm using weighted round-robin with health checks every 5 seconds. Internal east-west traffic between microservices traverses a dedicated VLAN (VLAN 200) on the Cisco Nexus fabric, which provides up to 400 Gbps of aggregate switching capacity. A critical dependency that complicates the migration is our integration with three on-premises systems operated by Vietnamese government agencies: the General Department of Taxation's e-invoice API (accessible only via a dedicated MPLS circuit from Viettel), the Social Insurance system's data feed (delivered as nightly SFTP batches to our DMZ), and the State Bank of Vietnam's payment verification endpoint (accessible via a VPN tunnel with IP whitelisting). These integrations cannot be simply lifted and shifted to the cloud; they require either maintaining on-premises bridge servers or negotiating new connectivity arrangements with each agency, which historically takes 6-12 months. Our network team, led by Senior Network Engineer Pham Van Khanh, has proposed a hybrid connectivity architecture using AWS Direct Connect from the existing data center to the AWS ap-southeast-1 region, with a VPN backup connection to the ap-southeast-5 (Hanoi) region. This approach maintains the existing government integrations while routing all other traffic through the cloud. The estimated monthly cost for this hybrid connectivity is \$3,200 (Direct Connect) plus \$450 (VPN), compared to the current \$1,800 for the two fiber connections alone.

2.3 Application Portfolio Analysis

The Engineering team conducted a comprehensive application assessment using the Gartner 6R framework (Rehost, Replatform, Repurchase, Refactor, Retire, Retain) to categorize each application and determine the optimal migration strategy. This assessment involved 48 person-days of effort across the Platform, Backend, and DevOps teams, and the results were validated by external consultants from Deloitte Vietnam. The assessment revealed several applications that are tightly coupled and must be migrated together as a group, which we have designated as 'migration units'. The most complex migration unit is the ChatAssist platform, which consists of 7 microservices, 3 databases (PostgreSQL primary, Redis cache, Elasticsearch), and 2 ML model serving instances. These components share a service mesh (Istio) and rely on sub-millisecond latency for inter-service communication, meaning they cannot be split across on-premises and cloud during migration without significant performance degradation. The second most complex unit is the Finance Platform, which includes the ERP integration layer, the e-invoice system,

and the payroll processing engine. This unit has the government agency dependencies described in Section 2.2, plus additional complexity from the fact that the payroll engine uses a proprietary library from Vietnamese vendor HRM Pro that is licensed to specific hardware MAC addresses. Re-licensing for cloud instances requires a contract amendment that the Procurement team has been negotiating since November 2025. The full application portfolio breakdown is as follows: Rehost (lift-and-shift): 8 applications including monitoring dashboards, CI/CD runners, development environments, and internal wikis. Replatform: 5 applications that will be containerized and deployed to EKS, including the customer portal, partner API gateway, and notification service. Refactor: 3 applications that require significant architectural changes, including ChatAssist (moving to event-driven architecture), the reporting engine (moving to serverless), and the data pipeline (moving to managed services). Retire: 2 legacy applications (the old PHP admin panel and the deprecated v1 API) that will be decommissioned. Retain: 2 applications that will remain on-premises permanently (the government integration bridge servers and the physical access control system).

3. Target Architecture

3.1 AWS Account Structure and Landing Zone

The target AWS architecture follows the AWS Well-Architected Framework and implements a multi-account strategy using AWS Organizations. The account structure has been designed to enforce security boundaries, cost allocation, and blast radius containment. At the top level, we have a Management Account that contains only AWS Organizations, SSO configuration, and consolidated billing. Below this, we have four Organizational Units (OUs): Security OU (containing the Log Archive and Security Tooling accounts), Infrastructure OU (containing the Network Hub and Shared Services accounts), Workloads OU (containing separate Production, Staging, and Development accounts for each product line), and Sandbox OU (containing individual developer sandbox accounts with spending limits). All accounts are provisioned using AWS Control Tower with custom Service Control Policies (SCPs) that enforce our security baseline. Key SCPs include: denying access to regions outside ap-southeast-1 and ap-southeast-5, requiring encryption at rest for all S3 buckets and EBS volumes, preventing the creation of IAM users (all human access must go through AWS SSO with SAML federation to our Okta tenant), and requiring VPC flow logs to be enabled in all VPCs. The landing zone was designed by our Cloud Architecture team in collaboration with the AWS Professional Services team over a 6-week engagement that concluded in January 2026.

3.2 Networking and Connectivity

The cloud networking architecture centers on a Transit Gateway in the ap-southeast-1 region that serves as the hub for all network connectivity. The Transit Gateway connects to: the production VPCs for each product line (ChatAssist VPC: 10.100.0.0/16, Customer Portal VPC: 10.101.0.0/16, Internal Tools VPC: 10.102.0.0/16), the shared services VPC (10.200.0.0/16) which hosts the EKS control plane, container registry, and monitoring infrastructure, and the Direct Connect gateway for hybrid connectivity back to the on-premises data center. Inter-region connectivity to the ap-southeast-5 (Hanoi) region uses Transit Gateway peering with a dedicated peering attachment for the data replication VPC. This ensures that data residency requirements are met by replicating specific data categories to the Hanoi region while keeping the primary compute in Singapore for latency optimization (our user base is 60% in Ho Chi Minh City, 25% in Hanoi, and 15% elsewhere in Vietnam). Each product VPC follows a consistent subnet design with public subnets (for ALBs only), private application subnets (for EKS worker nodes and application instances), private data subnets (for RDS, ElastiCache, and OpenSearch), and isolated subnets (for Lambda functions and batch processing). Security groups follow a zero-trust model where each microservice has its own security group with rules explicitly allowing only the required ports and source

security groups, rather than CIDR-based rules.

3.3 Kubernetes Architecture (EKS)

All containerized workloads will run on Amazon EKS with managed node groups using a mix of instance types optimized for each workload category. The cluster architecture uses a single EKS cluster per environment (production, staging, development) with namespace-based isolation for each product line. We chose single-cluster over multi-cluster to reduce operational overhead, but we enforce hard multi-tenancy boundaries using: Kubernetes RBAC with OIDC identity federation from Okta, NetworkPolicies enforced by Calico CNI plugin, ResourceQuotas and LimitRanges per namespace, and OPA Gatekeeper policies for admission control. The production cluster is configured with three managed node groups: a general-purpose group using m6i.2xlarge instances (8 vCPU, 32 GB RAM) with cluster autoscaler scaling between 6 and 24 nodes, a memory-optimized group using r6i.2xlarge instances (8 vCPU, 64 GB RAM) for Elasticsearch and Redis workloads scaling between 3 and 12 nodes, and a GPU-optimized group using g5.2xlarge instances (1 NVIDIA A10G GPU, 8 vCPU, 32 GB RAM) for ML model serving scaling between 2 and 8 nodes. Spot instances are used for the staging environment and for production batch processing jobs (data pipeline, report generation, ML training) to reduce costs by approximately 65%. The cluster uses AWS Load Balancer Controller for ingress, with separate Application Load Balancers for external-facing services and internal-only ALBs for service-to-service communication that requires HTTP-level routing. Service mesh functionality is provided by AWS App Mesh, which we chose over Istio to reduce the operational complexity of managing a separate control plane.

4. Migration Phases

4.1 Phase 1: Foundation and Internal Tools (Months 1-3)

Phase 1 focuses on establishing the cloud foundation and migrating low-risk internal tools. The first month is dedicated entirely to landing zone deployment, network connectivity setup, and security baseline configuration. The engineering team will deploy the Transit Gateway, establish the Direct Connect circuit (which has a 4-week provisioning lead time from the AWS side), configure the VPN backup, and validate end-to-end connectivity by running a suite of network tests from both the on-premises environment and the cloud. During this month, the Security team will also deploy their monitoring stack: AWS CloudTrail logs flowing to the centralized Log Archive account, Amazon GuardDuty enabled in all accounts, AWS Config rules for compliance monitoring, and Datadog agents deployed to the shared services infrastructure. In months 2 and 3, we will migrate the following internal tools: Confluence wiki (rehosted to EC2), Jenkins CI/CD runners (replatformed to EKS with self-hosted GitHub Actions runners), Grafana and Prometheus monitoring (rehosted to EKS with persistent EBS volumes), the internal project management tool (rehosted to EC2), and development environments (migrated to AWS Cloud9 and remote VS Code Server instances). These tools were selected for Phase 1 because they have no external user impact, minimal data sensitivity (no customer PII), and can be rolled back quickly if issues arise. The success criteria for Phase 1 are: all migrated tools operational in the cloud with less than 1 hour of cumulative downtime during migration, Direct Connect circuit operational with measured latency below 5ms (Singapore region) and below 15ms (Hanoi region), all security monitoring tools operational and alerting correctly, and cost tracking dashboards showing actual spend within 15% of the projected budget. The Phase 1 budget is \$45,000 for AWS infrastructure costs, \$12,000 for Direct Connect provisioning, and \$25,000 for the Deloitte advisory engagement.

4.2 Phase 2: Replatform Services (Months 4-8)

Phase 2 tackles the services that need containerization before moving to the cloud. This phase is the most engineering-intensive, as five applications need to be refactored from VM-based deployments to

containerized deployments on EKS. The Customer Portal is the largest of these applications, consisting of a React frontend served by Nginx, a Node.js BFF (Backend for Frontend) layer, and integration with three backend APIs. The current deployment process involves SSH-ing into two application servers and running a custom bash script that pulls the latest build artifact from Nexus and restarts the Nginx and Node processes. This will be replaced with a GitOps workflow using ArgoCD, where each merge to the main branch triggers a GitHub Actions pipeline that builds a Docker image, pushes it to ECR, updates the Kubernetes manifest in the GitOps repo, and ArgoCD automatically syncs the change to the cluster. The Partner API Gateway migration is particularly complex because it currently uses Kong API Gateway deployed on bare metal with custom Lua plugins for authentication, rate limiting, and request transformation. We have two options: migrate Kong to EKS (maintaining the custom plugins) or replace it with Amazon API Gateway with Lambda authorizers. After a 2-week proof-of-concept, the team recommended migrating Kong to EKS because the custom Lua plugins contain significant business logic that would take an estimated 6 weeks to rewrite as Lambda functions, and the API Gateway approach would introduce additional latency due to the Lambda cold start problem for infrequently called endpoints. The Notification Service migration will leverage AWS SES for email delivery (replacing our self-hosted Postfix server), Amazon SNS for push notifications (replacing Firebase Cloud Messaging direct integration), and Amazon SQS for the message queue (replacing RabbitMQ). This is a replatform rather than a refactor because the service's architecture remains largely the same; we are simply replacing self-managed middleware with managed AWS services.

4.3 Phase 3: Data Migration (Months 6-12)

Phase 3 overlaps with Phase 2 and focuses on migrating databases and data stores to managed AWS services. This is the highest-risk phase of the entire migration because data loss or corruption would directly impact customers and could have legal consequences under Vietnamese data protection regulations. The migration team has developed a detailed runbook for each database migration, with mandatory sign-offs from the Database Administrator (Nguyen Van Binh), the application team lead, and the DPO (Tran Minh Duc) before each migration can proceed. The primary PostgreSQL database (currently running PostgreSQL 15.4 on a dedicated server with 128 GB RAM and NVMe storage) will be migrated to Amazon RDS for PostgreSQL Multi-AZ in the ap-southeast-1 region, with a read replica in the ap-southeast-5 (Hanoi) region for data residency compliance. The migration approach uses AWS Database Migration Service (DMS) for the initial full load followed by continuous replication (CDC) to keep the target in sync during the parallel-run period. The estimated full load time for our 2.3 TB database is approximately 18 hours over the Direct Connect link, and we plan to execute this during a weekend maintenance window. After the full load, DMS CDC will maintain synchronization with an expected replication lag of less than 5 seconds under normal load. The cutover plan involves: stopping all write traffic to the source database (by putting the application in read-only mode), waiting for the CDC replication lag to reach zero, updating the application connection strings to point to the RDS endpoint, and resuming write traffic. The total expected downtime for the database cutover is 15-30 minutes. The Redis cache migration is straightforward: since the cache is ephemeral, we will simply provision an Amazon ElastiCache for Redis cluster with the same configuration (3 shards, 2 replicas per shard, r6g.xlarge node type) and update the application configuration. The cache will be cold after migration, so we expect a temporary increase in database load and API latency for approximately 2 hours as the cache warms up. The Elasticsearch migration to Amazon OpenSearch Service is the most complex data migration due to the size of our search indices (approximately 800 GB across 12 indices) and the requirement to maintain search functionality during migration. We will use the snapshot and restore approach: take a snapshot of the existing Elasticsearch cluster, store it in S3, and restore it to the OpenSearch cluster. During the snapshot transfer, which takes approximately 4 hours, search results may be slightly stale but the service will remain available.

4.4 Phase 4: Production Workloads (Months 13-18)

The final phase migrates the crown jewels: the ChatAssist production platform and customer-facing APIs. This phase cannot begin until all Phase 3 data migrations are complete and have been running stably in the cloud for at least 30 days. The ChatAssist migration is architecturally the most complex because it involves simultaneously migrating 7 tightly coupled microservices, 2 ML model serving instances, and the real-time WebSocket infrastructure that supports 50,000 concurrent connections at peak. The migration strategy for ChatAssist uses a canary deployment approach: we will route 1% of traffic to the cloud version initially, monitor for 48 hours, then progressively increase to 5%, 10%, 25%, 50%, and finally 100% over a 3-week period. Each traffic increase requires explicit approval from the Engineering Manager and must pass automated quality gates: error rate below 0.1%, p99 latency below 500ms, ML model prediction accuracy within 2% of the on-premises baseline, and zero data consistency errors detected by the reconciliation job. The traffic routing will be controlled at the DNS level using AWS Route 53 weighted routing policies, with the Palo Alto firewall maintaining the on-premises endpoint and the AWS ALB serving the cloud endpoint.

5. Cost Analysis and Financial Planning

The financial model for the cloud migration was developed jointly by the Finance team and the Cloud Architecture team, with validation from AWS's pricing specialists. The model uses a 5-year total cost of ownership (TCO) comparison between three scenarios: Scenario A (status quo with data center lease renewal at the new rate), Scenario B (the proposed hybrid cloud migration), and Scenario C (a full cloud migration with no on-premises components). After extensive analysis, the steering committee selected Scenario B because it offers the best balance of cost savings (42% TCO reduction vs. Scenario A), risk mitigation (maintaining on-premises connectivity for government integrations), and implementation timeline (18 months vs. 24 months for Scenario C). The year-by-year projected costs for Scenario B are: Year 1 (migration year): \$580,000 total (\$280,000 AWS infrastructure during ramp-up, \$120,000 migration tooling and services, \$80,000 training and certification, \$60,000 on-premises data center costs during overlap, \$40,000 contingency). Year 2: \$420,000 (\$350,000 AWS infrastructure at full scale, \$40,000 residual on-premises costs for government bridges, \$30,000 optimization and tooling). Year 3: \$380,000 (\$340,000 AWS with reserved instance savings, \$40,000 on-premises). Year 4: \$360,000 (\$320,000 AWS with further optimization, \$40,000 on-premises). Year 5: \$350,000 (\$310,000 AWS, \$40,000 on-premises). The cumulative 5-year TCO for Scenario B is \$2,090,000, compared to \$3,600,000 for Scenario A (including the new lease terms, hardware refresh cycles, and staffing for on-premises operations), representing a savings of \$1,510,000 or 42%. The break-even point occurs in Month 19, one month after the migration is projected to be complete. The model assumes a 3% annual AWS price decrease (conservative based on historical trends) and a 5% annual increase in on-premises costs (inflation, maintenance contracts, and power costs). Sensitivity analysis shows that even if AWS costs are 20% higher than projected, the migration still achieves a 28% TCO reduction over 5 years.